

INDEPENDENT PEER-REVIEW PACKET

EC-021

**Uniform Positive Typed Finite-Q4 Palm Mass of
the Fixed-q Corridor**

Cold Review Packet 15 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	15 of 20
Object ID	EC-021
Object class	Uniform finite-dimensional probability lemma
Exact statement	There exist $r_1 > 0$ and $c_{Q4} > 0$ such that $\inf_{\{0 < r \leq r_1\}} P_{\{r, Q4\}}^{\{MS\}}(P01-UNIFORMQ-D4-BOX-001) \geq c_{Q4} > 0$.
Internal terminal label	CLOSED - EXACT UNIFORM POSITIVE TYPED FINITE-Q4 PALM MASS FOR SUFFICIENTLY SMALL r
Terminal date	2026-07-22
Recommended expertise	Gaussian probability / Palm laws / asymptotic uniformity

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Uniform Gaussian density lower bound, compact box, determinant/type lower bound, r^2 numerator and normalizer cancellation, and independent review.
Prior review record	Independent Q1-Q10 review passed; the EC-020 deterministic dependency was already terminal.
Binding exclusions	No exact full-field mass, exact selected-branch adjacency, numerical c_{Q4} or maximal r_1 , P0.1/P0.2, or Theorem B.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.

- Object-specific exclusions: No exact full-field mass, exact selected-branch adjacency, numerical c_{Q4} or maximal r_1 , $P0.1/P0.2$, or Theorem B.

4. Object-specific hostile checks

- Verify the density lower bound is uniform on one common small- r interval.
- Check the endpoint determinant/type lower bound and the full normalizer upper bound.
- Audit the r^2/r^2 cancellation and denominator direction.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

TERMINAL STATUS NOTICE — 2026-07-23

Dylan M. Roy approved EC-021 through HA-008. GP-CLS-141-v1.0 is the terminal closure record. The historical package body below is preserved unchanged as the pre-approval decision surface. EC-021 is terminal and removed from active attention; exact full-field and downstream theorem exclusions remain binding.

[[STATUS:TERMINAL]] [[OBJ:GP-CLS-141-v1.0]] [[QUEUE:EC-021-REMOVED]]

GP-CLS-PROP-025-v1.0 — CONDITIONAL TERMINAL CLOSURE PACKAGE
UNIFORM POSITIVE TYPED FINITE-Q4 PALM MASS OF THE FIXED-q CORRIDOR

Artifact ID: GP-CLS-PROP-025-v1.0

Prepared by: GP — OpenAI GPT-5.6 Thinking

Created: 2026-07-22

Class: CLS-PROP — complete additive package pending Dylan M. Roy's exact terminal approval

Authority: none before human approval

Canonical impact: NONE before approval; afterward only the exact narrow finite-Q4 probability object below

Queue candidate: EC-021

[[OBJ:GP-CLS-PROP-025-v1.0]] [[QUEUE:EC-021]] [[OBJ:CL-AUD-074-v1.0]] [[STATUS:HUMAN-APPROVAL-READY]]
[[STATUS:NOT-TERMINAL]]

1. EXACT OBJECT PROPOSED FOR CLOSURE

Let $B_q = P01-UNIFORMQ-D4-BOX-001$ and let $P_{\{r, Q4\}}^{\{MS\}}$ be the typed finite-Q4 Palm law defined from:

- the exact normalized-periodized Gaussian law γ_r of the nine midpoint jets;
- the normalized quartic endpoint Hessians $H_M^{(4)}, H_S^{(4)}$;
- the finite-Q4 determinant/type weight $W_{\{r, Q4\}}$;
- the finite-Q4 normalizer $Z_{\{r, Q4\}} = E_{\{\gamma_r\}}[W_{\{r, Q4\}}]$.

There exist $r_1 > 0$ and $c_{Q4} > 0$ such that

$$\inf_{\{0 < r \leq r_1\}} P_{\{r, Q4\}}^{\{MS\}}(B_q) \geq c_{Q4} > 0.$$

Because EC-020 supplies B_q subset $H_r^{(4)}$ on the same deterministic interval, this also gives uniform positive finite-Q4 degree-four capture-event mass for sufficiently small r .

This package closes only that qualitative finite-dimensional finite-Q4 statement. No numerical value of r_1 or c_{Q4} is adopted.

2. PRIMARY EVIDENCE

- GP-DER-116-v1.0 — governing uniform finite-Q4 mass proof.
- GP-AUD-117-v1.0 — same-line adversarial audit and mutation analysis.
- GP-DER-043 — typed finite-Q4 Palm law, quartic endpoint weight, and r^2 normalizer factorization.
- GP-DER-044 — exact-torus nine-jet Gaussian regularity and compact-density argument.

- - GP-DER-114-v1.0 / GP-DATA-114-v1.1 / GP-AUD-115-v1.0 — fixed-q corridor dependency.
- - GP-AUD-112-v1.0 — mandatory separation of finite-Q4 and exact full-field Palm objects.
- - GP-PRP-132-v1.0 — frozen Q1–Q10 review surface.
- - GP-CLS-PROP-024-v1.0 / CL-AUD-073-v1.0 — reviewed EC-020 deterministic dependency.

3. INDEPENDENT REVIEW

CL-AUD-074-v1.0 performs a line-independent Q1–Q10 proof review and verifies:

- - every probability statement uses $P_{\{r,Q4\}^{\{MS\}}}$, finite-Q4 endpoint Hessians, and $Z_{\{r,Q4\}}$;
- - no exact-field Palm weight is substituted;
- - small- r nondegeneracy/continuity gives a compact positive Gaussian density minimum $c_{\phi}>0$;
- - integration is in physical q coordinates with fixed q -width and no hidden $q=rs$ Jacobian;
- - the EC-020 inclusion is the correct deterministic dependency and was reviewed concurrently in CL-AUD-073;
- - strict quartic type stability on B_q ;
- - rigorous determinant-factor bounds $d_M, d_S \geq 0.979930952835$ and $w_0 \geq 0.96026$;
- - $W_{\{r,Q4\}} \geq r^2 w_0$;
- - $Z_{\{r,Q4\}} = r^2 z_r$ with z_r uniformly bounded above on $0 < r \leq r_1$;
- - exact r^2/r^2 cancellation;
- - $\text{Vol}(B_q) = 1.024e-18$;
- - $r_1 = \min(r_{GR}, 1/20)$, with no numerical r_1 or c_{Q4} claimed;
- - all exact-field, A_r , $P0.1/P0.2$, theorem, machine, ballot, and release exclusions.

CL-AUD-074 records one non-material precision correction: use $d_M, d_S \geq 0.979930952835$ or a shorter conservative bound, rather than the optimistic final trailing digits appearing in the GP prose. This package adopts the conservative reviewed values.

CL-AUD-074 ends with:

APPROVE EC-021 EXACT UNIFORM FINITE-Q4 SCOPE

The declared line-independence gate is therefore satisfied, conditional on the concurrently reviewed EC-020 dependency. EC-020 is now also human-approval-ready under GP-CLS-PROP-024-v1.0.

4. MATHEMATICAL BASIS

On B_q and $0 < r \leq r_1$:

- - γ_r has density $\phi_r \geq c_{\phi}>0$;
- - $\text{Vol}(B_q) > 0$;
- - $W_{\{r,Q4\}} \geq r^2 w_0$ with $w_0 > 0$;
- - $Z_{\{r,Q4\}} \leq r^2 C_Z$ with $C_Z < \infty$.

Therefore

$$\begin{aligned}
 & P_{\{r, Q4\}^{\{MS\}}}(B_q) \\
 &= \int_{B_q} W_{\{r,Q4\}}(u) \phi_r(u) du / Z_{\{r,Q4\}} \\
 &\geq r^2 w_0 c_{\phi} \text{Vol}(B_q) / (r^2 C_Z) \\
 &= w_0 c_{\phi} \text{Vol}(B_q) / C_Z \\
 &=: c_{Q4} > 0.
 \end{aligned}$$

All constants in the final expression are independent of r on the declared interval.

5. AFFECTED DEPENDENCIES UPON TERMINAL APPROVAL

Retire from active attention:

- - repeated derivation of the finite-Q4 r^2 numerator/normalizer cancellation;
- - whether B_q has uniformly positive mass under $P_{\{r,Q4\}}^{\{MS\}}$ for sufficiently small r ;
- - whether a compact density lower bound, quartic weight lower bound, and normalizer upper bound suffice for the qualitative finite-Q4 result;
- - HB-032's narrow EC-021 proof-review request;
- - the EC-021 review portion of GP-PRP-132.

Record as available finite-dimensional input:

- - a uniform positive finite-Q4 degree-four corridor-mass component for later exact-field architecture, without transferring its probability to the exact full field.

6. UNAFFECTED DEPENDENCIES AND EXCLUSIONS

This closure does not prove or close:

- - exact endpoint-Hessian control;
- - exact full-field Palm mass;
- - exact selected-branch adjacency A_r ;
- - a numerical r_1 , c_{Q4} , or c_A ;
- - exact-field uniform support or remainder probability;
- - P0.1, P0.2, Theorem B, elder-rule pairing, or lifetime laws;
- - equality between finite-Q4 and exact full-field normalizers;
- - theorem promotion, GP-AUTO-034 activation, machine replacement, ballot passage, release, deletion, or destructive action.

Failure of an exact-field bridge does not invalidate this separate finite-Q4 probability statement.

7. PROPOSED TERMINAL LABEL

CLOSED — EXACT UNIFORM POSITIVE TYPED FINITE-Q4 PALM MASS OF THE FIXED-q DEGREE-FOUR CORRIDOR FOR SUFFICIENTLY SMALL r

8. REQUIRED REGISTER AND LEDGER UPDATES AFTER HUMAN APPROVAL

1. Append EC-021 to the Closure Log with the exact finite-Q4 law, proof chain, conservative determinant bounds, CL-AUD-074 review, EC-020 dependency, exclusions, terminal label, and reopening rule.
2. Mark EC-021 **TERMINAL — REMOVED FROM ACTIVE ATTENTION** in the Easy Closure Queue.
3. Mark HB-032's narrow finite-Q4 proof-review request **SATISFIED / RETIRED**.
4. Update Dashboard terminal count and next technical item.
5. Add Evidence Lineage entries for GP-DER-116, GP-DER-043/044, EC-020 evidence, and CL-AUD-074.
6. Add Review Independence disclosure identifying the CL line, its independent computations, and its **CANNOT-VERIFY** list.
7. Add a Transition Register entry changing only the EC-021 finite-Q4 component from reviewed candidate to terminal narrow closure.
8. Preserve all earlier finite-Q4, exact-field separation, correction, and review artifacts.

9. HUMAN APPROVAL REQUIREMENT

Dylan M. Roy must explicitly state:

APPROVE EC-021 EXACT UNIFORM FINITE-Q4 SCOPE

General authorization to continue Drive work, organize files, or pursue research does not substitute for this item-level terminal decision.

10. REOPENING RULE

Reopen only by additive evidence showing one or more of:

- - hidden substitution of the exact full-field Palm object;
- - singular or discontinuous nine-jet law invalidating the compact density argument;
- - incorrect physical-coordinate measure or hidden shrinking fixed-s width;
- - failure of the EC-020 set inclusion on the declared shared domain;
- - nonpositive quartic determinant/type factor on B_q ;
- - incorrect r^2 numerator or normalizer factorization;
- - lack of a finite uniform C_Z or positive c_ϕ ;
- - scope contamination or object mismatch.

Failure of exact-field transfer, P0.1, P0.2, or any downstream theorem does not reopen this finite-Q4 closure unless it contradicts the exact finite-dimensional statement itself.

11. CURRENT STATUS

PACKAGE: COMPLETE.

PRIMARY EVIDENCE: COMPLETE.

NON-GP REVIEW: PASS — CL-AUD-074-v1.0.

EC-020 DEPENDENCY REVIEW: PASS — CL-AUD-073-v1.0; terminal human decision still open.

HUMAN APPROVAL: OPEN.

TERMINAL: NO.

END GP-CLS-PROP-025-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT UNIFORM POSITIVE TYPED FINITE-Q4 PALM MASS FOR SUFFICIENTLY SMALL r
Closure date	2026-07-22
Evidence basis	Uniform Gaussian density lower bound, compact box, determinant/type lower bound, r^2 numerator and normalizer cancellation, and independent review.
Independent review already recorded	Independent Q1-Q10 review passed; the EC-020 deterministic dependency was already terminal.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No exact full-field mass, exact selected-branch adjacency, numerical c_{Q4} or maximal r_1 , $P_{0.1}/P_{0.2}$, or Theorem B.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____

Reproduction artifacts

Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1ZwSNFhPjPUBEPxT6ySMmKLI8yTtcsN9XMSmBNpIwasQ/edit>

Closure / terminal record:

https://docs.google.com/document/d/1Z8XWIS5gVvkiZL67sgx8Ca71LR2HFxU_96HULcuOCMI/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-025-v1.0 — Conditional Closure of Uniform Positive Finite-Q4 Palm Mass

Local source snapshot SHA-256 (DOCX export, transport checksum only):
6cafe1dd0d16173d02731e7f2795390f6d303a4511f481489540fe0b55a8a42b

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.